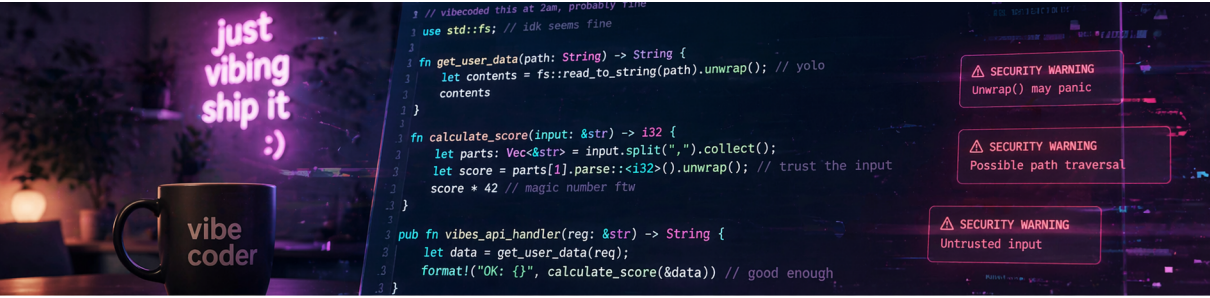
universität innsbruck

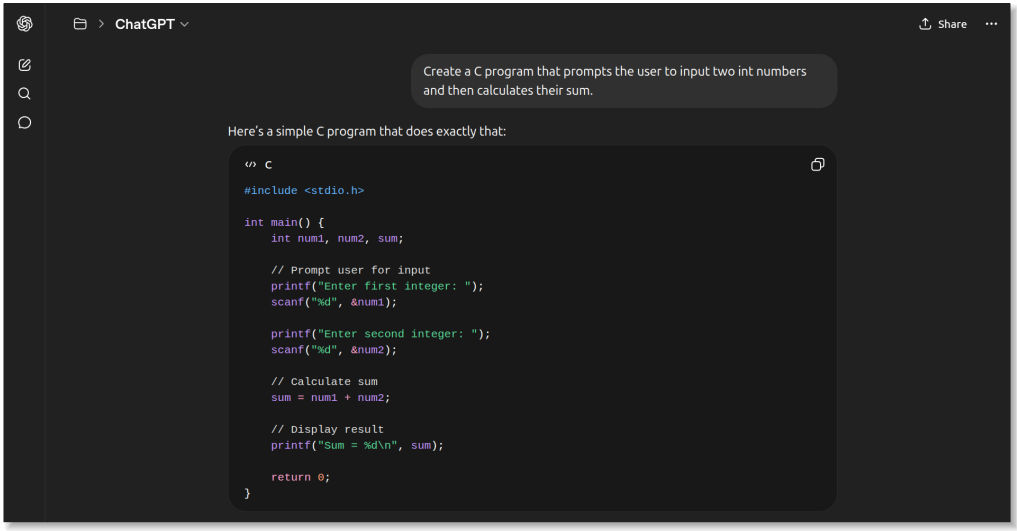


Vibe Coding and Software Vulnerabilities

Context Presentation: Vulnerabilities introduced by LLMs

Stefan Wagner

Motivation



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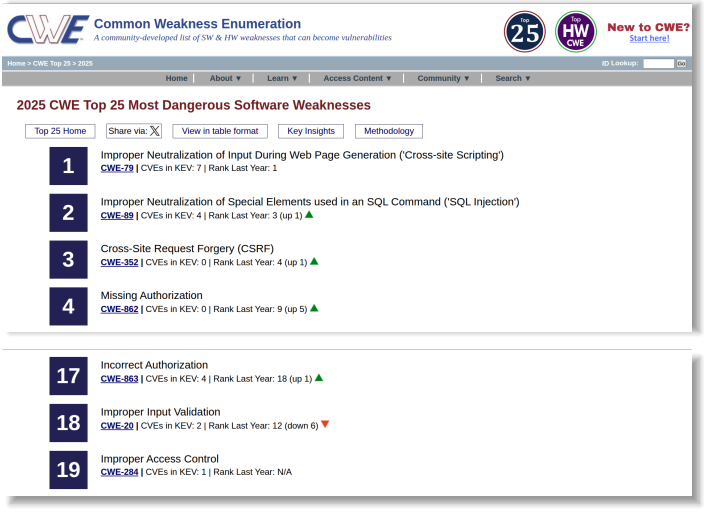

MITRE: Common Weakness Enumeration (CWE) · `https://cwe.mitre.org/top25/archive/2025/2025_cwe_top25.html`

Vibe Coding & Software Vulnerabilities

Observation

- LLMs are now part of everyday programming
- Generated code can contain vulnerabilities
- Vulnerabilities may be missed during review
- Suggested fixes can be incomplete

Open question

Does this generalize across models, prompts, tasks, and programming languages?

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⇒ **Overview of current research and implications for Rust**

Systematic Literature Review (SLR)

From Vulnerabilities to Remediation

- Analysis of 102 studies on LLMs in code security
- Focus on “recent“ work, mainly 2024 – 2025

From Vulnerabilities to Remediation: A Systematic Literature Review of LLMs in Code Security

ENNA BASIC, Department of Computer Science, Örebro University, Sweden and Epiroc Rock Drills, Sweden
ALBERTO GIARETTA, Department of Computer Science, Örebro University, Sweden

Large Language Models (LLMs) have emerged as powerful tools for automating programming tasks, including security-related ones. However, they can also introduce vulnerabilities during code generation, fail to detect existing vulnerabilities, or report nonexistent ones. This systematic literature review investigates the security benefits and drawbacks of using LLMs for code-related tasks. In particular, it focuses on the types of vulnerabilities introduced by LLMs when generating code. Moreover, it analyzes the capabilities of LLMs to detect and fix vulnerabilities, and examines how prompting strategies impact these tasks. Finally, it examines how data poisoning attacks impact LLMs performance in the aforementioned tasks.

CCS Concepts: • **Security and privacy** → **Software security engineering**; **Vulnerability scanners**; • **Computing methodologies** → *Artificial intelligence*.

Additional Key Words and Phrases: Large Language Models, LLMs, Security Vulnerabilities, LLM-generated Code, Vulnerability Detection, Vulnerability Fixing, Prompt Engineering, Data Poisoning

Enna Basic and Alberto Giaretta. *ACM Comput. Surv.* 1, 1 (March 2026), 41 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Large Language Models (LLMs) are Machine Learning (ML) models designed for natural language processing [12]. Representative models such as ChatGPT [3], Llama [106] and GitHub Copilot [4] gained extreme popularity and recognition in the last couple of years due to their ability to perform well across different tasks. Throughout this paper, we use the term "LLM" to refer to large autoregressive Transformer language models that support prompt-driven, open-ended code generation and analysis. In contrast, we use Pre-trained Language Model (PLM) to refer to pre-trained (often smaller) encoder-only Transformer models (e.g., CodeBERT, GraphCodeBERT) that are typically fine-tuned for task-specific prediction or classification rather than open-ended generation. We treat these PLMs and other task-specific neural models as traditional ML baselines and include them only when they serve as comparison points for LLM-based approaches.

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Basic & Giaretta · Preprint 2026 [2]

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Key contribution

- Identifies common vulnerability categories
- Highlights differences in experimental setups

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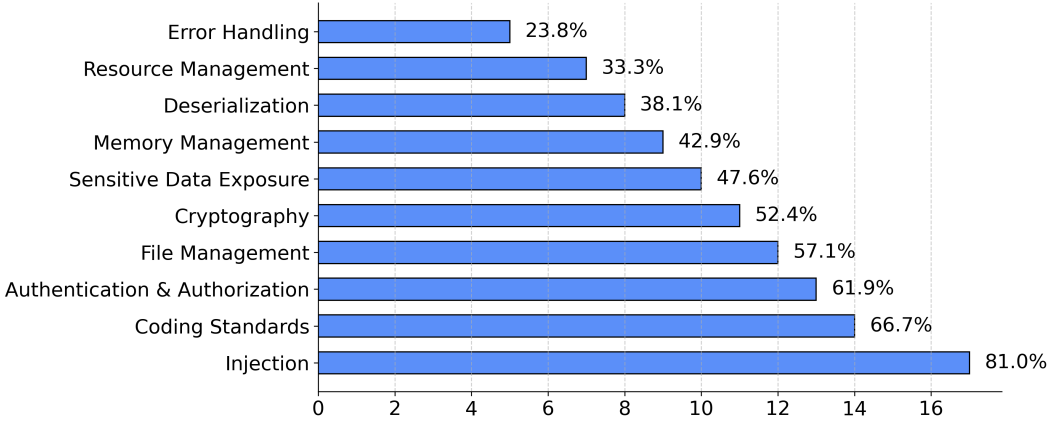
SLR: Vulnerability Categories in LLM-Generated Code



Number of studies reporting each category (N=21) [2]

SLR: Differences in Experimental Setups

Research Challenges

- Studies use different setups, tasks, and evaluation methods
- Direct comparison across papers is difficult

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⇒ **Studies can be grouped by their evaluation approaches:**

Humans vs. LLMs	Controlled Code Generation
Existing Code Analysis	Mitigation & Fine-Tuning

Evaluation Approach: Humans vs. LLMs

⇒ **Do LLMs lead to more insecure code than human developers?**

- User studies (2023) [3, 4]
- StackOverflow comparison (2024) [5, 6]

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Findings

- No clear winner
- Humans remain a major source of vulnerabilities
- LLMs influence how developers write and review code

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Limitations

- Studies may not generalize: small and task-specific
- Results depend strongly on user behavior and interaction with the model

Evaluation Approach: Controlled Code Generation

⇒ **How secure is LLM-generated code under controlled conditions?**

- Self-defined tasks (2022–2024) [7, 8, 9, 10, 11]
- Dataset / benchmark evaluations (2022–2025) [12, 13, 14, 15, 16, 17, 18, 19]

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- Differences between models exist
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Limitations

- Artificial tasks targeted at specific CWEs
- No iterative, developer-driven coding ⇒ Not “Vibe Coding“

Evaluation Approach: Existing Code Analysis

⇒ **How secure is LLM-generated code in real-world usage?**

- DevGPT dataset: Shared ChatGPT conversations (2024) [20]
- GitHub projects: AI-labeled code (2025) [21]

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Limitations

- Selection bias ⇒ Most code is not marked as LLM-generated
- Limited insight into development and review process

Evaluation Approach: Mitigation & Fine-Tuning

⇒ **Can LLMs be trained to generate more secure code?**

- Guided code generation through prefix tuning (2023) [22]
- Fine-tuning on secure vs. insecure code (2024) [23]

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Limitations

- Requires modified or fine-tuned models
- Evaluated under controlled conditions

Research Gaps & Limitations: Vibe Coding

Research Gaps

- Existing studies do not reflect modern LLM usage
- Focus on small, isolated tasks: function / file level
- No iterative, multi-round interaction with LLMs

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Open Question

Is Vibe Coding safe?

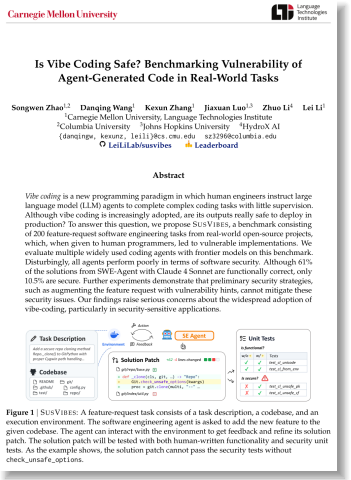
Research Gaps & Limitations: Vibe Coding

Is Vibe Coding Safe?

- Agent-based code generation
- Real-world programming tasks
- Multi-step interaction with execution feedback

Key Findings

- ∼83% of working solutions are insecure
- Stronger models ≠ more secure code
- Secure prompting ≠ more secure code



Carnegie Mellon University

Language Technologies Institute

Is Vibe Coding Safe? Benchmarking Vulnerability of Agent-Generated Code in Real-World Tasks

Songwen Zhao[1,2] Danqing Wang[1] Kexun Zhang[1] Jiaxuan Luo[1,3] Zhuo Li[4] Lei Li[1]
[1]Carnegie Mellon University, Language Technologies Institute
[2]Columbia University [3]Johns Hopkins University [4]HydroX AI
{danqingw, kexunz, lei1i}@cs.cmu.edu sz3296@columbia.edu

LeiLiLab/susvibes Leaderboard

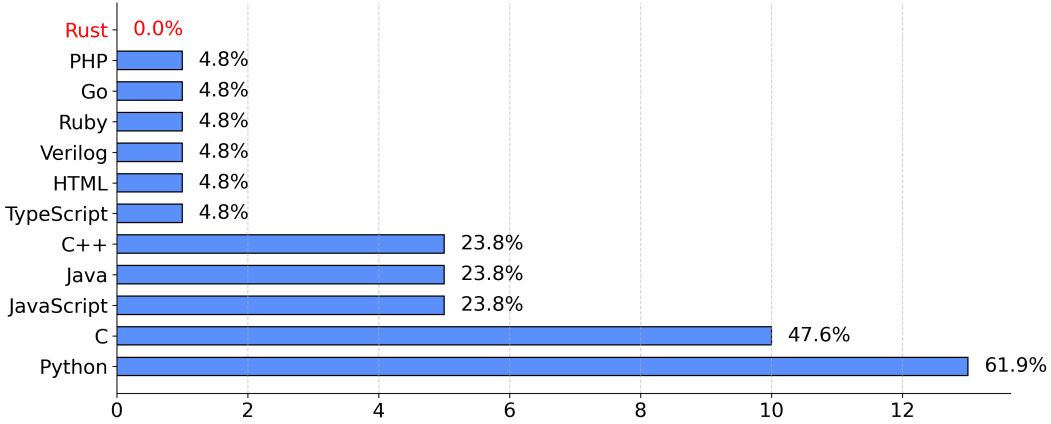
Abstract

Vibe coding is a new programming paradigm in which human engineers instruct large language model (LLM) agents to complete complex coding tasks with little supervision. Although vibe coding is increasingly adopted, are its outputs really safe to deploy in production? To answer this question, we propose SusVibes, a benchmark consisting of 200 feature-request software engineering tasks from real-world open-source projects, which, when given to human programmers, led to vulnerable implementations. We evaluate multiple widely used coding agents with frontier models on this benchmark. Disturbingly, all agents perform poorly in terms of software security. Although 61% of the solutions from SWE-Agent with Claude 4 Sonnet are functionally correct, only 10.5% are secure. Further experiments demonstrate that preliminary security strategies, such as augmenting the feature request with vulnerability hints, cannot mitigate these security issues. Our findings raise serious concerns about the widespread adoption of vibe-coding, particularly in security-sensitive applications.

Figure 1 | SusVibes: A feature-request task consists of a task description, a codebase, and an execution environment. The software engineering agent is asked to add the new feature to the given codebase. The agent can interact with the environment to get feedback and refine its solution patch. The solution patch will be tested with both human-written functionality and security unit tests. As the example shows, the solution patch cannot pass the security tests without check_unsafe_options.

Zhao et al. · Preprint 2026 [24]

Research Gaps & Limitations: Rust



Programming Language Distribution across studies (N=21) [2]

Research Gaps & Limitations: Rust

Observation

- Existing studies focus on `Python`, `C`, `JavaScript`, `C++`, `Java`
- Rust is not part of the evaluated language distribution

Why Rust matters

- Prevents many memory-safety vulnerabilities by design
- LLMs may generate non-compiling or use unsafe patterns
- Vulnerabilities can still occur at higher levels

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Research Gap

No empirical evidence on LLM-generated vulnerabilities in Rust

Experiment Proposal

Approach

- Adapt controlled code-generation setup from Liu et al. (2024) [10]
- Define multiple Rust-specific tasks for selected CWEs
- Evaluate outputs from two LLMs

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Goal

Determine in which contexts LLMs generate vulnerable Rust code.

Summary

- LLMs generate code that can be vulnerable
- Four main evaluation approaches:

Humans vs. LLMs	Controlled Code Generation
Existing Code Analysis	Mitigation & Fine-Tuning

- “Vibe Coding“ has not really been studied yet
- Rust is missing in current evaluations

Experiment Goal

Determine in which contexts LLMs generate vulnerable Rust code.

Experiment Proposal: Core Idea

⇒ **Two classes of vulnerabilities**

Prevented by Rust

- use-after-free
- null pointer dereference
- out-of-bounds access
- buffer overflow

Questions:

- Does the code compile?
- Are unsafe patterns used?

Not prevented by Rust

- integer overflow
- OS command injection
- path traversal
- missing input validation

Questions:

- Do vulnerabilities still appear?
- Does Rust reduce them?

Unsafe Rust Example

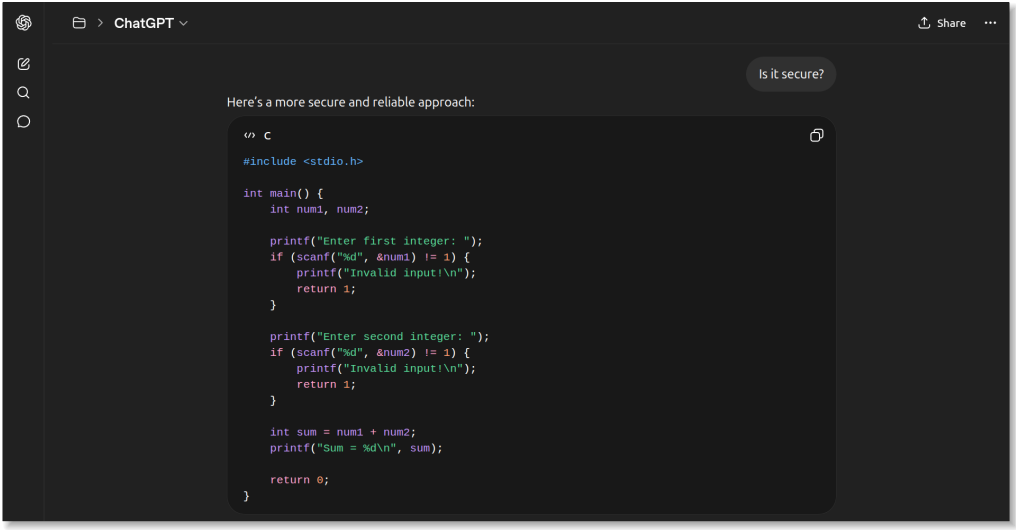
unsafe.rs

```rust
use std::rc::Rc;

fn main() {
    let raw_ptr: *const String;

    {
        let secret = Rc::new(String::from("SECRET: database_password_123"));
        raw_ptr = Rc::as_ptr(&secret);

        println!("{}", secret);
    } // secret dropped here -> memory freed

    // println!("{}", secret); prevented by Rust

    unsafe {
        // Undefined behavior: may print garbage / crash / or still look correct
        let secret = Rc::from_raw(raw_ptr);
        println!("{}", secret);
    }
}
```

DEBUG CONSOLE OUTPUT PROBLEMS TERMINAL PORTS

```
sw@5tefan543:~/unsafe_rust_example$ rustc unsafe.rs
sw@5tefan543:~/unsafe_rust_example$ ./unsafe
SECRET: database_password_123
Gj~E??XD7S?_password_123
```

Vulnerability Detection & Fixing



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